# SAURABH PATEL

**Electronics & Communication Engineering | Embedded Systems | Robotics**

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LinkedIn | GitHub | Portfolio

## Professional Summary

Electronics and Communication Engineering student at NIT Nagaland with hands-on project experience in embedded systems, sensor integration, wireless communication, robotics, UAVs, and AI-assisted screening. Interested in embedded hardware, device-level development, prototyping, and electronics engineering internships.

## Education

**National Institute of Technology Nagaland** Aug 2024 – Jun 2028
B.Tech. in Electronics and Communication Engineering

**St. Xavier School, Bhiwadi** 2021 – 2022
Class XII – Physics, Chemistry and Mathematics

**St. Xavier School, Bhiwadi** 2019 – 2020
Class X

## Technical Skills

- **Programming:** C, Embedded C, Python, MySQL
- **Microcontrollers & Platforms:** ESP32, STM32, Arduino, Raspberry Pi
- **Embedded Systems:** Sensor interfacing, GPIO, GPS integration, motor control, hardware debugging, circuit prototyping, soldering
- **Communication:** UART, $I^2C$, SPI, Wi-Fi, Bluetooth, ESP-NOW
- **Robotics & AI:** IMU integration, multirotor systems, flight controllers, Edge Impulse, image classification
- **Tools:** Arduino IDE, MATLAB, Git, GitHub, VS Code

## Major Projects

**MineSafe-X – Low-Visibility Mining Vehicle Safety System** *ESP32, sensors, wireless communication*
- Developing a retrofit driver-assistance prototype for mining vehicles operating in fog and low visibility.
- Integrates multi-sensor distance monitoring, vehicle-to-vehicle communication, tracking, and collision-warning concepts.

**Aerial Fire Detection & Suppression Drone** *STM32, ESP32, Raspberry Pi, Edge Impulse*
- Developing an aerial prototype for AI-assisted fire recognition, GPS/IMU-based sensing, and payload deployment.
- Explores onboard image classification and embedded control.

**Ostereo-AI – Osteoarthritis Risk Marker Screening System** *AI, gait analysis*
- AI-assisted screening workflow combining questionnaires, walking-video/gait analysis, and knee X-ray inputs to support risk assessment and referral.

**AC Line Fault Detection System**
- Prototype concept for detecting and indicating AC line fault conditions through electronic sensing and indication.

**Additional Projects:** Multirotor UAV, ultrasonic obstacle detection, Bluetooth-controlled RC car.

## Experience

**Electronics Manufacturing Intern** | Elexoplus India Private Limited, Bhiwadi, Rajasthan 21 June – 19 July 2026
- Gained practical exposure to electronics manufacturing, component assembly, production workflows, and system testing.

**Sanchar Mitra** | Department of Telecommunications (DoT) August 2025 – Present
- Supporting telecom awareness activities and public engagement initiatives.

## Workshops & Training

**Workshop on Assembly of Obstacle Avoidance Based on Multirotor UAV** | NIELIT September 2025
- Hands-on workshop on assembling a multirotor UAV with an obstacle-avoidance system; certificate of participation received.

## Achievements

- Runner-Up, Code For Nagaland 26-Hour Hackathon, Wildcard Track – Open Innovation, Team Ostereo-AI, October 2026.
- Invited Exhibitor, TXG Expo Nagaland, Aerial Recognition project, 2026.
- Winner, Robo GP, Technical Board, NIT Nagaland, March 2025.

## Leadership & Positions of Responsibility

- Robotics Secretary, Robotics Club, NIT Nagaland August 2026 – Present
- Joint Secretary, Entrepreneurship & Incubation Cell, NIT Nagaland
- Internship Coordinator, NIT Nagaland August 2026 – Present
- Campus Ambassador, The Polaris Fellowship September 2026 – Present
- Assistant Club Secretary, Robotics August 2025 – August 2026

## Languages

Hindi | English | Haryanvi